Skills Summary

Integration by Parts: Choosing the Pieces

Use this reference while completing your worksheet or when studying.

Skill 1 — One Pass: Choosing u and dv

Formula. $\int u\,dv = uv - \int v\,du.$

Choosing. Take as u the factor that gets *simpler* when differentiated (a power of x, a logarithm, an inverse trig function), and as dv the factor you can integrate without making things worse (an exponential, a sine or cosine, or just dx).

Example: Evaluate $\int xe^{3x}dx$.

Step 1. Differentiating x removes it while the exponential integrates back to itself, so x is the factor to call u.

Step 2. $u = x$, $v = \frac{e^{3x}}{3}$, so the integral is $\frac{xe^{3x}}{3} - \frac{1}{3}\int e^{3x}dx$.

$\Rightarrow \quad \boldsymbol{\frac{xe^{3x}}{3} - \frac{e^{3x}}{9} + C}$

Try it: Evaluate $\int xe^{4x}dx$.

check: $\frac{xe^{4x}}{4} - \frac{e^{4x}}{16} + C$

Skill 2 — Logarithms and Inverse Trig

Why they go on top. $\ln x$ and $\arctan x$ have no elementary antiderivative you already know, but their *derivatives* are simple algebraic fractions. So they are always u, never dv.

When a logarithm is the only factor, take $dv = dx$ and $v = x$: parts still applies with nothing else in sight.

Example: Evaluate $\int x^2 \ln x\,dx$.

Step 1. The logarithm cannot be integrated directly but differentiates to $1/x$, so it must be u and the power must be dv.

Step 2. $v = \frac{x^3}{3}$, so the integral is $\frac{x^3 \ln x}{3} - \frac{1}{3}\int x^2 dx$.

$\Rightarrow \quad \boldsymbol{\frac{x^3 \ln x}{3} - \frac{x^3}{9} + C}$

Try it: Evaluate $\int x^3 \ln x\,dx$.

check: $\frac{x^4 \ln x}{4} - \frac{x^4}{16} + C$

Skill 3 — Repeating Until It Closes

Two ways a second pass ends it. With x^n times an exponential or a trig function, each pass lowers the power by one until the polynomial is gone. With $e^x \sin x$ the integral *returns* after two passes — call it I, collect, and solve for I algebraically.

Keep the same kind of factor as dv on both passes, or the second pass undoes the first.

Example: Evaluate $\displaystyle\int x^2 \sin x \, dx$.

Step 1. One pass drops the power from two to one but leaves the same shape, so plan on two passes with dv trigonometric both times.
Step 2. First pass gives $-x^2 \cos x + 2\int x \cos x \, dx$, and the second gives $2\,(x \sin x + \cos x)$.
$\Rightarrow$ $\boldsymbol{-x^2 \cos x + 2x \sin x + 2 \cos x + C}$

Try it: Evaluate $\displaystyle\int x^2 \cos x \, dx$.

check: $\boldsymbol{x^2 \sin x + 2x \cos x - 2 \sin x + C}$